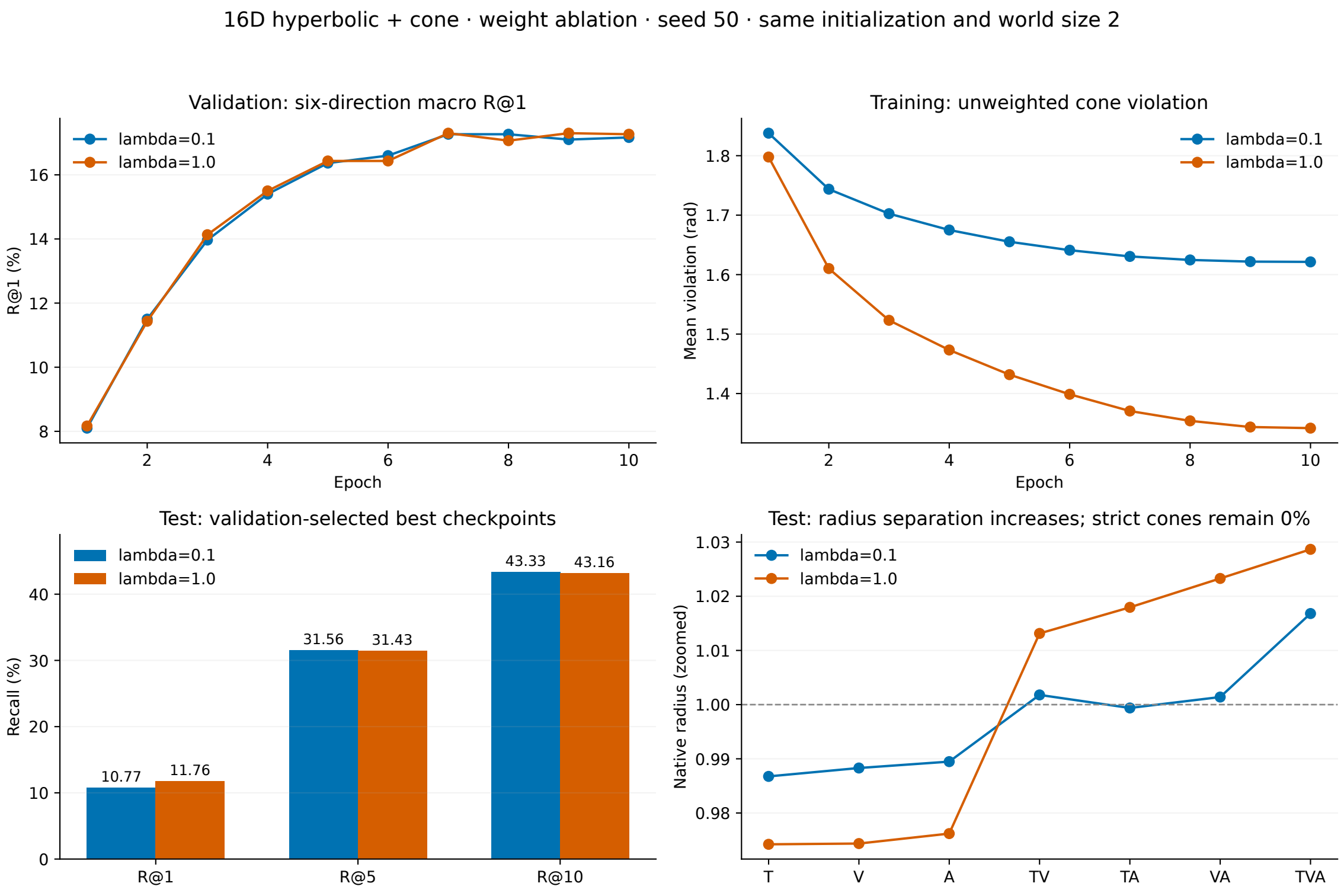
16D hyperbolic + cone · weight ablation · seed 50 · same initialization and world size 2
Validation: six-direction macro R@1
lambda=0.1
lambda=1.0
R@1 (%)
Epoch
Training: unweighted cone violation
Mean violation (rad)
Test: validation-selected best checkpoints
Recall (%)
10.77
11.76
31.56
31.43
43.33
43.16
R@1
R@5
R@10
Test: radius separation increases; strict cones remain 0%
Native radius (zoomed)
T
V
A
TV
TA
VA
TVA